INCH-POUND

KK-I-570D
May 29, 1990
SUPERSEDING
KK-I-570C
November 11, 1975

FEDERAL SPECIFICATION

INSOLE, FOOTWEAR, LEATHER, CATTLEHIDE

This specification is approved by the Commissioner, Federal Supply Service, General Services Adminstration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers insoles cut from cattlehide leather.

1.2 Classification.

1.2.1 Types, tannages, and classes. The insoles shall be of the following types, tannages, and classes, as specified (see 6.2):

Type I - Full grain or full grain which has been degrained or buffed (combination tanned):

Tannage a - Chrome retan
Tannage b - Veteable retan

Type II - Full grain or full grain which has been degrained or buffed (straight vegetable tanned)

Type III - Flexible splits (combination tanned):

Tannage a - Chrome retan
Tannage b - Vegetable retan

Class 1 - Men's
Class 2 - Women's

Benneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8335

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2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications:

A-A-203	-	Paper, Kraft, Untreated
PPP-B-636	-	Boxes, Shipping, Fiberboard

Federal Standards:

FED-STD-123	-	Marking for Shipment (Civil Agencies)
FED-STD-311	-	Leather, Methods of Sampling and Testing

(Activities outside the Federal Government may obtain copies of Federal specications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes, are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Fransisco, CA; Los Angeles, CA; and Seattle, WA).

(Federal Government activities may obtain copies of Federal Specifications Standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies).

Military Specifications:

MIL-L-35078	-	Loads, Unit; Preparation of Semiperishable Subsistence Items; Clothing, Personnal Equipment and Equipage; General Specification For

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members For Containers and Pallets

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(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Types I and III. The leather for types I and III insoles shall be cattlehide tanned by one of the following methods, as specified (see 6.2):

Tannage a - Tanned by a chrome-tanning process followed by a vegetable retannage
Tannage b - Tanned by a vegetable-tanning process followed by a chrome retannage

3.2.2 Type II. The leather for type II insoles shall be cattlehide, tanned with only vegetable-tanning materials.

3.3 Chemical requirements.

3.3.1 Types I and III. The leather for types I and III insoles shall conform to the chemical requirements in table I, when tested as specified in 4.4.1.2.

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TABLE I. Chemical requirements, types I and III

Characteristic	Minimum	Maximum
Water soluble material (percent) 1/	---	26.0
Chloroform soluble material (percent) 1/	1.8	10.0
Insoluble ash (percent) 1/:		
Type I	---	7.0
Type III	---	10.0
pH value	3.0	4.0
Chromic oxide (percent) 1/	2.0	---
Shrinkage temperature	90°C	---
Paranitrophenol 2/	0.18	0.70

1/ Calculated on the moisture-free basis.

2/ Unless otherwise specified (see 6.2).

3.3.2 Type II. The leather for type II insoles shall conform to the chemical requirements listed in table II, when tested as specified in 4.4.1.2.

TABLE II. Chemical requirements, type II

Characteristic	Minimum	Maximum
Water soluble material (percent) 1/	---	35.0
Chloroform soluble material (percent) 1/	3.0	10.0
Insoluble ash (percent) 1/	---	1.0
pH value	3.0	4.0
Paranitrophenol 2/	0.18	0.70

1/ Calculated on the moisture-free basis.

2/ Unless otherwise specified (see 6.2).

3.4 Physical requirements. The leather for insoles shall conform to the following physical requirements.

3.4.1 Area stability. The leather shall not shrink more than 10 percent in area when tested as specified in 4.4.1.2.

3.4.2 Cracking (types I and II). The leather shall not crack when tested as specified in 4.4.1.2.

3.4.3 Piping (types I and II). The leather shall not pipe when tested as specified in 4.4.1.2.

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3.4.4 Water absorption. The leather shall absorb not less than 0.50 grams of water per specimen when tested as specified in 4.4.1.2.

3.5 Thickness. The thickness of the insoles shall be as specified (see 6.2). A plus tolerance of 1 iron shall be permitted and no minus tolerance shall be allowed.

3.6 Finish.

3.6.1 Types I and II. The leather for types I and II insoles shall be full grain or full grain leather which has been degrained or buffed. The flesh side of the leather shall be smooth and free from loose fibers. When full grain leather is used, the grain side of the insoles shall be degrained or buffed.

3.6.2 Type III. The leather for type III insoles shall be the first split under the grain surface and shall be that type commercially known as flexible splits. Both sides shall be smooth and free from loose fibers.

3.7 Patterns. The outline of insoles shall be equal to or greater than the outline of the paper insole patterns furnished by the Government.

3.8 Workmanship. The insoles shall conform to the quality and grade of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2.1 and 4.4.2.2 and tested for the characteristics specified in 4.4.1.2.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the insole conforms to the requirements specified in 3.2.1 and 3.2.2.

4.4.1.2 Component testing of cattlehides. The methods of testing specified in FED-STD-311, wherever applicable, and as listed in table III shall be followed except that pieces for physical and chemical tests shall be taken from each sample unit in the sample size for tests. Specimens for chemical tests shall be taken not closer than 1/2 inch to any edge. The sample unit shall be one 8 by 12 inch swatch of leather, and the sample size shall be 15 swatches cut from 15 separate hides randomly selected from throughout the lot to be tested. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any of the following conditions exist:

a. More than one failure occurs for shrinkage temperature.

b. More than two test failures occur for the remaining requirements applicable to the sample unit.

c. Any composite value fails to meet the specified requirements.

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TABLE III. Component testing

Characteristic	Requirement paragraph	Test method	Requirement applicable to Sample unit	Composite
Material identification	3.2.1 and 3.2.2	1/	- -	- -
Tannage	3.2.1 and 3.2.2	1/	- -	- -
Paranitrophenol	3.3.2	6711	- -	X
Water soluble material	3.3.1 and 3.3.2	6341	- -	X
Chloroform soluble material	3.3.1 and 3.3.2	6341	- -	X
Insoluble ash	3.3.1 and 3.3.2	6341	- -	X
pH value	3.3.1 and 3.3.2	6621	- -	X
Chromic oxide	3.3.1	6515	- -	X
Shrinkage temperature	3.3.1	7011	X	- -
Area stability	3.4.1	7031	X	- -
Cracking (types I and II)	3.4.2	4021 2/	X	- -
Piping (types I and II)	3.4.3	4111 2/	X	- -
Water absorption	3.4.4	8151 3/	X	- -

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The diameter of the mandrel shall be as follows:

a. 4 irons or less - 1/2 inch
b. 4-1/2 irons to 5-1/2 irons - 3/4 inch
c. 6 irons and over - 1 inch

NOTE: This test shall be conducted in accordance with the "Atmospheric Conditions for Testing" as specified in FED-STD-311. Leather for flesh split insoles for the tropical boot direct molded sole need not be tested for cracking and piping.

3/ Leather for flesh split insoles for the tropical boot direct molded sole need not be tested for water absorption.

4.4.2 End item examination. The defects found during these examinations shall be in accordance with the tables shown in 4.4.2.1 and 4.4.2.2. The lot size for these examinations shall be expressed in units of insoles. The sample unit for these examinations shall be one insole. Any sample unit containing one or more defects shall be classified as a defective unit. The sampling procedure for visual and dimensional examinations shall be in accordance with FED-STD-311, except that a sample size of 15 and a rejection number of 2 shall be applicable for the dimensional examination on lots consisting of 51 or more units of product. Separate examinations shall be performed for visual and dimensional characteristics. The sample for dimensional examination shall be obtained from the sample used for the visual examination.

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4.4.2.1 End item visual examination.

TABLE IV. End item visual defects

Examine	Defect
Brand	Brand on shank or forepart Heavy brand in heel area
Wrinkles	Heavy fat wrinkles
Grain damage	Grain damage which will not buff out Any deep cut or scratch Extensive cracked grain in shank or forepart Any pit
Holes	Any hole except pinhole
Scar	Any healed scar in channel area 2/ 4/
Hardness	Hard and bony
Slaughter cut	Deep cut Any cut parallel to and within the channel area 2/ 4/ Any cut parallel to and within the stitching area 1/ 4/ Penetration more than 1/3 the thickness of the bottom layer 3/
Pattern damage	Sole outline not conforming to required pattern
Fiber characteristics	Hard and very tight Loose and spongy Loose fibers on either side of insole (bottom layer) 3/
Finish	Type I or II not degrained or not buffed full grain Insoles excessively skived or leveled Type III not flexible split, i.e. first split under grain surface; sides not smooth and free from loose fibers Types I and II flesh side not smooth and free from loose fibers

1/ Applicable to tropical boot direct molded sole construction.

2/ Not applicable to direct molded sole construction.

3/ Applicable to bottom layer of tropical boot direct molded sole construction.

4/ Not applicable to stuck-on rib construction.

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4.4.2.2 End item dimensional examination.

TABLE V. End item dimensional defects

Examine	Defect
Size and area	Outline of insole not equal to or greater than outline of applicable insole pattern
Thickness	Thickness of insole not as specified 1/

1/ Thickness shall be determined by the wedge type gage, in accordance with Method 1011 of FED-STD-311. No individual thickness measurement shall be less than the minimum or greater than the maximum specified.

4.4.3 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.4 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation. Sixty pairs of insoles of one description only shall be evenly stacked and then securely tied with cotton tape or twine. When specified (see 6.2), ten pairs of insoles of one description shall be completely wrapped in 30 pound minimum basis weight kraft paper conforming to A-A-203. The wrap shall be secured with cotton tape or twine, or pressure-sensitive tape.

5.1.2 Commercial preservation. Insoles shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Insoles of one description only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each container packed with unwrapped insoles shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3. The weight of contents of each shipping container shall not exceed 65 pounds. Shipping contianers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2).

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5.2.2 Level B packing. Insoles of one description only, preserved as specified in 5.1, shall be packed in snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Each container packed with unwrapped insoles shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2.1 Water-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification, except that the inspection shall be in accordance with 4.4.3.

5.2.3 Commercial packing. Insoles preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), insoles, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallets shall be fabricated from wood groups I, II, III or IV of MIL-STD-731. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The insoles are used in military and civil agency footwear.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, tannage, and class (see 1.2.1 and 3.2.1).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. When the fungicide paranitrophenol is not required (see 3.3.1 and 3.3.2).

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e. Thickness required (see 3.5).
f. Levels of preservation and packing (see 5.1 and 5.2).
g. Whether ten pairs of insoles per package are desired (see 5.1.1).
h. Type and class of unit load required (see 5.2.1).
i. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
j. When palletization is required (see 5.3).
k. Type of marking required (see 5.4).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Full grain
Shoes
Tannage

MILITARY INTERESTS:

Custodians:

Army - GL
Navy - NU

Review activities:

Army - MD
DLA - CT

User activity:

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing activity:

Army - GL

(Project 8335-0169)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	KK-I-570D	1990 May 29

3. DOCUMENT TITLE
INSOLE, FOOTWEAR, LEATHER, CATTLEHIDE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
	(1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-651-5221	256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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